

Subsystem Specification Simulation & test

Let the whole system be exercised, failed on purpose and rehearsed at home, first against a simulated boat and then with more and more real hardware in the loop, with the same software throughout.

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Issue	Issue E (for review)
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Status	For review by the project owner
Parents	SRS Issue D, ADD Issue C, ICD Issue B
Content	3 allocated SRS requirements → 30 subsystem requirements (23 M, 7 S, 0 C)

Document control

Issue	Date	Change	By
A	28 September 2026	First issue, for review.	Claude (drafted)
B	28 September 2026	Hardware-in-the-loop rigs L1-L3 added (SIM-D13 to D24). Test catalogue rebuilt with IDs and extended with FMEA-derived scenarios (BOATY-FMEA-001).	Claude, owner request
C	28 September 2026	L1-10 (adult PIN) and B-07 (cell acceptance) added for FMEA FM-48 and A-20.	Claude, owner decision (CR-03)
D	29 September 2026	Simulator built (slices 1-2): our own boat model through SITL's JSON interface (ADD DD-20), B1 router topology, real services on a simulated Pi Zero. SIM-D01/02/07 updated; SIM-D25 to D28 added; test results shown against the catalogue; SC-02 at 3 s (CR-05).	Claude, owner decisions
E	29 September 2026	Slice 3: Mission Control in the loop. SC-41 to SC-43 added; SIM-D29 and D30 added; SC-31 and SC-33 evidence sources named.	Claude

Role	Name	Signature / date
Author	Claude, for Project Boaty	
Reviewer / approver	Project owner	

Review guidance: section 5 is what will be built and tested. Check that each requirement is right, sufficient and testable. Section 4 shows every parent requirement is covered (the build script checks this).

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1. Scope

1.1 Purpose

Let the whole system be exercised, failed on purpose and rehearsed at home, first against a simulated boat and then with more and more real hardware in the loop, with the same software throughout.

1.2 Boundary

Inside this subsystem	Outside (and who owns it)
• ArduPilot Rover 4.7.1 SITL • Boat and battery model (JSON physics) • Radio-link relay and B1 router stand-in • Simulated camera for B2 • Scenario runner and test suites (software/tests) • CI configuration • Rig L1: real computers + SITL • Rig L2: iron bird (real helm and power train) • Rig L3: water-tank thrust and weed rig	• The software under test (MCN, MCP services) • Pool and lake trials (per-subsystem and system test procedures)

1.3 Breakdown

Item	Description
SIM-1 Simulator	SITL + boat model (software/boaty/sim) + launcher; services run from software/boaty/mcp
SIM-2 Test suites	Unit, contract, scenario, FMEA-derived, LLM evaluation
SIM-3 CI	GitHub Actions workflows
SIM-4 Rig L1	Real Pi 5 panel, real Pi Zero + camera, real Wi-Fi; SITL on a laptop, reaching the Pi Zero UART through a USB-serial adapter
SIM-5 Rig L2 'iron bird'	Real FC, firmware and parameters, power board with key, ESCs and motors, GNSS, beacon, on a board; driven by the real Mission Control
SIM-6 Rig L3 tank	One pod in a water tub on a load cell, with current logging and real pond weed

2. Applicable documents

Document	Use
BOATY-SRS-001 Issue D	Parent requirements
BOATY-ADD-001 Issue C	Allocation, decisions, budgets, early verification items
BOATY-ICD-001 Issue B	Interface definitions (all listed in section 3 apply in full)

3. Interfaces

The subsystem shall meet every interface below as defined in the ICD. 'Owner' means this subsystem maintains the definition.

Interface	Other end	Role	ICD
IF-15 Site data	Mission Control	Conforms	\$5
IF-21 Simulation link	Mission Control	Owner	\$5

4. Parent requirements

Primary (3): this subsystem is responsible for meeting and verifying these SRS requirements.

SRS	Pri	Requirement (abridged)	Met by
SWE-004	M	The whole system shall run in simulation (helm simulator and simulated camera) on the Pi 5 or a laptop, with Mission...	SIM-D01, SIM-D02, SIM-D03, SIM-D04, SIM-D25, SIM-D26, SIM-D29, SIM-D13
SWE-005	M	Each failsafe requirement (FS-001 to FS-013) shall have an automated simulation scenario.	SIM-D06, SIM-D07, SIM-D27, SIM-D28, SIM-D30
SWE-007	S	Unit tests (pytest), linting (ruff) and type checking (mypy) shall run automatically on every push.	SIM-D11

5. Subsystem requirements

Requirements are numbered SIM-Dnn. Pri, Ver and Stage follow SRS conventions (M/S/C; Test, Demonstration, Inspection, Analysis, Simulation; SIM, BENCH, POOL, LAKE). 'Trace' lists the parent SRS requirements, interfaces (IF), decisions (DD) and early verification items (V) it implements.

5.1 Simulator

ID	Requirement	Pri	Ver	Stage	Trace
SIM-D01	SITL shall run the same ArduPilot version as the helm (Rover 4.7.1) and the controlled parameter file, flying our own skid-steer boat model through the JSON interface (ADD DD-20).	M	D	SIM	SWE-004, V-12, DD-20
SIM-D02	The real B2/B3 code shall serve IF-03 in simulation, with a simulated camera rendering images at the boat's true position.	M	T	SIM	SWE-004, IF-21
SIM-D03	One command shall start the simulator on a laptop or the Pi 5 at a given site's home.	M	D	SIM	SWE-004
SIM-D04	Mission Control shall run against the simulator with configuration changes only.	M	D	SIM	SWE-004, SWE-003
SIM-D05	A rehearsal mode shall run the physical panel against the simulator for crew practice.	S	D	SIM	OPS-012

5.2 Tests

ID	Requirement	Pri	Ver	Stage	Trace
SIM-D06	Each FS-001 to FS-013 requirement shall have an automated scenario with explicit pass criteria (SC-01 to SC-13, section 6). FS-008 is proven on rig L2.	M	T	SIM	SWE-005
SIM-D07	Fault injection shall use SITL simulation parameters (SIM_GPS1_ENABLE, SIM_MAG1_ORIENT), boat-model faults (motor loss or reversal, weed drag, wind, battery drain, key out) and radio-link impairment (cut, loss, latency), mapped per scenario in the test code.	M	T	SIM	SWE-005, IF-21
SIM-D08	Contract tests for IF-03 and IF-14 shall run against both the simulator and real hardware.	M	T	SIM	IF-14, IF-03
SIM-D09	The validator's adversarial suite (≥ 50 cases) and a property-based comparison against an independent geometry implementation shall run in CI (SC-31).	M	T	SIM	VAL-006
SIM-D10	The LLM evaluation set (≥ 30 instructions, including adversarial child phrasing) shall run on demand with a per-run cost cap (SC-33).	S	T	SIM	NLI-003, NLI-006

5.3 CI and reproducibility

ID	Requirement	Pri	Ver	Stage	Trace
SIM-D11	Every push shall run ruff, mypy and unit tests. SITL scenarios shall run on pull requests and nightly.	S	I	SIM	SWE-007
SIM-D12	Tool and dependency versions (SITL commit, Python lock file) shall be pinned in the repository.	M	I	SIM	SWE-006

5.4 Built in slices 1-2 (Issue D)

ID	Requirement	Pri	Ver	Stage	Trace
SIM-D25	The boat model shall take its parameters from the Key Component List (mass, thrust ahead/astern, drag, windage, pack capacity and resistance) and be re-fitted to thrust-stand and pool data when they exist.	M	A	SIM	SWE-004, DD-20

ID	Requirement	Pri	Ver	Stage	Trace
SIM-D26	The simulated boat shall have the boat's MAVLink topology: SITL's port joined by a B1 router to the radio relay and to the services, so a mission-computer failure takes the bank link with it.	M	D	SIM	SWE-004, IF-04, IF-21
SIM-D27	Every timing criterion shall be measured in simulated time from truth (commanded outputs, thrust, position), not from telemetry round trips.	M	I	SIM	SWE-005
SIM-D28	Each run shall write per-scenario evidence (measured values against the criterion) and a report (software/results). Known native gaps are marked as such, never hidden.	M	I	SIM	SWE-005, SAF-007
SIM-D29	Mission Control shall run in the simulator as it runs on the Pi 5 (same code, simulated time), with Claude's replies from a stand-in client that produces the API's response format. The live API is exercised separately by the recorded evaluation set (SC-33).	M	I	SIM	SWE-004, IF-11
SIM-D30	Evidence that needs no simulator (validator coverage, property-based runs) shall be written to software/results/unit_evidence.json and reported with the SITL results.	S	I	SIM	VAL-006, SWE-005

5.5 Hardware-in-the-loop rigs (Issue B)

ID	Requirement	Pri	Ver	Stage	Trace
SIM-D13	Rig L1 shall connect SITL to the real Pi Zero 2W through a 3.3 V USB-serial adapter at 115200 baud on the Pi's UART, so the production router configuration is used unchanged. The real Pi 5 panel and Wi-Fi are used.	M	D	BENCH	SWE-004, IF-04, IF-01
SIM-D14	Rig L1 shall timestamp STOP from button edge to the SITL mode change with ≤ 10 ms resolution, over ≥ 100 trials (L1-03).	M	T	BENCH	FS-009, COM-003
SIM-D15	Rig L1 shall switch the Pi Zero's 5 V supply under script control for power-cut endurance testing (L1-08).	S	T	BENCH	IF-08
SIM-D16	Rig L1 shall allow the sealed box with the Pi Zero to be heat-soaked at 30 °C (heat lamp or sun) while running a mission (L1-09).	S	T	BENCH	PWR-010
SIM-D17	Rig L2 shall mount the real FC (flashed firmware and controlled parameters), power board with fuse and magnetic key, ESCs, motors, GNSS with sky view, and beacon, driven by the real Mission Control over the real link.	M	D	BENCH	SAF-001, IF-05, IF-06, IF-07, IF-09
SIM-D18	Rig L2 shall support physical fault injection: GNSS unplug and data-line cut, AP off, bench-supply voltage ramp, ESC signal cut, key removal, FC power-cycle, and thrust steps.	M	T	BENCH	FS-001, FS-002, FS-004, FS-008, MOD-003, PWR-005
SIM-D19	Rig L2 shall record motor current, 5 V rails and ESC signals (logic analyser or scope), and detect motor stop within 50 ms (ESC telemetry or optical).	M	T	BENCH	FS-008, FS-009, PWR-005
SIM-D20	Rig L2 shall run with props removed in air, or with pods submerged in a covered tank. Removing the key is the emergency stop, and the adult-only rules apply.	M	I	BENCH	OPS-007, CHD-003
SIM-D21	Rig L3 shall measure one pod's thrust (± 0.05 N) and current, forward and astern, at 0-100% in 10% steps (L3-01).	M	T	BENCH	NAV-001, ENV-002, IF-06
SIM-D22	Rig L3 shall introduce real pond weed at the inlet and record whether the weed-shedding routine clears it (L3-02).	S	T	BENCH	ENV-005, FS-006
SIM-D23	Every rig test shall be recorded with date, software and firmware versions, parameter hash and results, and committed to the repository.	M	I	BENCH	SWE-006
SIM-D24	Every FMEA failure mode with severity ≥ 8 shall be exercised by at least one SIM, rig or bench test in the catalogue. The build checks this.	M	A	SIM	SAF-006, FS-013

6. Test catalogue

Every test has an ID, which the FMEA (BOATY-FMEA-001) and the test procedures reference. SC = simulator scenario; L1/L2/L3 = rig tests; B = bench; R = rehearsal; P = pool. Injection parameter names are fixed per SITL version and kept in the scenario code. **Result** is read from software/results/sitl_results.json (slices 1-2); '-' means not yet built (mostly slice 3, Mission Control).

6.1 Failsafe scenarios (one per FS requirement)

ID	Test	Injection / method	Pass criterion	Refs	Result
SC-01	Battery drain to 35% then 15%	Simulated battery drain	RTL at 35%; alarm at 15%; reaches home	FS-001	Pass
SC-02	Link cut in STEERING	Drop UDP 14550	HOLD ≤ 3 s (CR-05); RTL at 10 s (B4)	FS-002	Pass
SC-03	Link cut in AUTO	Drop UDP 14550	Mission continues; RTL at 60 s (B4)	FS-003, FM-31	Pass
SC-04	GNSS failure 5 s then restore	SITL GPS failure	Motors stop ≤ 3 s; RTL after 10 s healthy (B6)	FS-004, FM-01, FM-06	Pass
SC-05	Heavy drag at throttle $\geq 50\%$	SITL drag / wind	HOLD ≤ 5 s plus event	FS-005, FM-16	Pass (via V-05; B7 backs up at 10 s)
SC-06	Drag released after burst 2	SITL + B5	≤ 3 bursts; resumes mission	FS-006, FM-27	Pass
SC-07	Kill all MCP services mid-mission	Stop stub and B-services	Mission completes; RTL; HOLD at home	FS-007, FM-26	Pass
SC-08	Helm signal loss to ESCs	Covered by L2-10	See L2-10	FS-008	-
SC-09	STOP in every state	Panel emulation	Motors stop ≤ 1 s	FS-009	Pass (panel emulation, 4 armed states)
SC-10	Moisture flagged	Stub health = moisture	RTL ≤ 2 s	FS-010, FM-24	Pass
SC-11	GNSS loss during battery RTL	Combined	Motors-stopped (HOLD) wins	FS-011	-
SC-12	Log and announce audit	Log inspection over SC-01 to SC-11	Every event logged, announced ≤ 2 s	FS-012	-
SC-13	Single-fault sweep across mission phases	Scripted matrix of SC-20 to SC-39 faults	Never leaves the fence under power	FS-013	-

6.2 FMEA-derived simulator scenarios

ID	Test	Injection / method	Pass criterion	Refs	Result
SC-20	GNSS glitch: 20-50 m jump for 2 s near the fence	SITL GPS glitch	No uncommanded exit; glitch rejected or HOLD	FM-02	-
SC-21	GNSS frozen: stale position while moving	SITL GPS freeze	Detected; motors stop ≤ 3 s	FM-03	-
SC-22	Compass offset 30° and 90°	SITL compass offset	Yaw fallback or HOLD; stays inside the fence	FM-04, FM-05	-
SC-23	Helm restart mid-mission	Restart SITL process	Motors stop; boots disarmed; alarm at Mission Control	FM-07	-
SC-24	Parameter differs from baseline	Change one failsafe parameter	Arming blocked; difference shown	FM-09	Pass
SC-25	Fence missing or disabled	Skip the fence upload	Arming refused	FM-10	Pass

ID	Test	Injection / method	Pass criterion	Refs	Result
SC-26	Site file lat/lon swapped; wrong site	Corrupt the site file	Linter and pre-arm both refuse	FM-11	Pass
SC-27	Breach on the far side of the island	Push the boat out with wind	RTL path avoids the exclusion	FM-13	-
SC-28	Persistent breach (wind pushing out)	Strong offshore wind	Motors stop by 30 s / 10 m	FM-14	Pass
SC-29	Single motor failure	Zero one output	Divergence watchdog → HOLD ≤ 20 s	FM-17	Pass
SC-30	Boat-service command fuzz	Random forbidden commands	All blocked by the filter	FM-27	Pass (unit fuzz, 6,000 cases)
SC-31	Validator adversarial + property-based	Hypothesis vs independent geometry	No accepted mission crosses a boundary	FM-34	Pass (400 generated, 112 accepted, closest 2.999 m)
SC-32	Read-back corruption	Alter one item in transfer	GO stays disabled	FM-35, FM-47	Pass
SC-33	LLM evaluation with adversarial phrasing	Recorded instruction set	Schema-valid; must-decline declined	FM-36, FM-37	Not run: needs the API key (to ols/nli_eval.py, 33 cases ready)
SC-34	Link degradation: 30/60/90% loss, 2 s latency	Network impairment	FS-002/003 behaviour holds; STOP latency logged	FM-39, FM-40	-
SC-35	Home sanity	Arm with GNSS unsettled / far from site home	Refused or warned before GO	FM-46	Pass
SC-36	Interrupted upload	Drop the link mid-transfer	No partial mission accepted	FM-47	-
SC-37	Foreign GCS heartbeat	Second system-255 source	C7 refuses to operate; alarm	FM-41	Pass
SC-38	First-motion heading check	Compass reversed	HOLD ≤ 10 s after start	FM-05, FM-18	Pass
SC-39	Capacity overstated by 30%	Wrong capacity parameter	Voltage backstop triggers RTL in time	FM-15	-
SC-40	Arm lands in HOLD	Arm with default RC input present	Armed in HOLD, motors off	FM-53	Pass
SC-41	Instruction to captain's log	Typed instruction, stand-in Claude reply, full Mission Control	Validated, read back equal, flown inside the fence, home, auto-disarm at 60 s, photos sha256-verified before ack	SWE-004, VAL-010, MCN-D47	Pass
SC-42	Persistent breach with B7 absent	Gale; boat services not running	C1 stops motors before 30 s or 10 m outside	FM-14, MCN-D59	Pass
SC-43	Boat stops itself	Dead motor mid-mission	Reason spoken ≤ 2 s after HOLD; resume needs the PIN	MCN-D60	Pass

6.3 Rig L1: real computers, simulated boat

ID	Test	Injection / method	Pass criterion	Refs	Result
L1-01	End-to-end mission, real computers	Voice → captain's log over real Wi-Fi and UART	Completes; all artefacts logged	SWE-004	-
L1-02	Panel self-test and state × button table	Real buttons	Matches ICD IF-12	FM-32, FM-33	-
L1-03	STOP latency, 100 trials	Button edge → SITL mode	$P99 \leq 0.5 \text{ s}$	FM-40, FS-009	-
L1-04	Photo sync, 200 photos	Real Wi-Fi at 10 m	$\leq 5 \text{ min}$; hashes verified	CAM-007	-
L1-05	UART load	Maximum telemetry + services	GCS heartbeat jitter $< 0.5 \text{ s}$	FM-28	-
L1-06	Mission Control power pulled mid-mission	Unplug the Pi 5	B4 RTL at 60 s	FM-31	-
L1-07	Range walk on land	Walk the Pi Zero away	RSSI vs distance; flap behaviour as SC-34	FM-39	-
L1-08	Pi Zero power-cut ×50	Switched 5 V	No filesystem damage	FM-26	-
L1-09	Sealed-box heat soak 30 °C	Heat lamp	No throttling for 60 min	PWR-010	-
L1-10	Adult PIN: unlock, timeout, lockout	Web UI	10 min unlock; 5 failures → 5 min lockout; motors still need the key	FM-48, MC-008	-

6.4 Rig L2: iron bird

ID	Test	Injection / method	Pass criterion	Refs	Result
L2-01	Motor direction and channel map	Props off	Left/right and sense correct	FM-18	-
L2-02	Arming gated on the key	Key in/out	Refused with key out; rail $< 0.5 \text{ V}$	FM-22, MOD-003	-
L2-03	GNSS unplugged	Pull connector	EKF failsafe HOLD	FM-01	-
L2-04	GNSS data line cut, power on	Cut TX	Detected $\leq 3 \text{ s}$	FM-03	-
L2-05	Compass vs motor current	Motor-interference calibration	Interference $\leq 30\%$	FM-04	-
L2-06	Heading vs reference	Known bearing	Within 10°	FM-05	-
L2-07	FC power-cycle with motors running	Interrupt FC supply	Motors stop; boots disarmed	FM-07	-
L2-08	Thrust-step brownout	Full reverse → forward	5 V rails in regulation	FM-08, PWR-005	-
L2-09	Supply voltage ramp	Bench supply	Battery failsafe at thresholds	FM-15	-
L2-10	ESC signal cut and restore	Cut the signal lead	Stop $\leq 1 \text{ s}$; restart only from neutral	FM-19, FM-20, FS-008	-
L2-11	STOP to motors stopped	Panel STOP	$\leq 1 \text{ s}$	FS-009, FM-40	-
L2-12	Beacon patterns	Cycle states	Each state distinguishable	REC-004	-
L2-13	Key-switch short detection	Bridge the MOSFET	Checklist rail test flags it	FM-22	-

6.5 Rig L3: water tank

ID	Test	Injection / method	Pass criterion	Refs	Result
L3-01	Thrust and current curves	Load cell, 0-100%	$\geq 2 \text{ N}$ forward per pod; current $\leq 8 \text{ A}$	NAV-001, IF-06	-

ID	Test	Injection / method	Pass criterion	Refs	Result
L3-02	Weed fouling trials	Real pond weed	Cleared in ≥ 2 of 3	FM-16, ENV-005	-
L3-03	Cruise power estimate	Thrust vs drag model	≤ 10 W at 1.0 m/s	PWR-009	-

6.6 Bench, rehearsal and pool items referenced by the FMEA

ID	Kind	Test	Pass criterion	Refs
B-01	BENCH	Prop guard probe	No blade contact	FM-21, FM-45
B-02	BENCH	BMS protection	BMS trips; fuse intact	FM-23
B-03	BENCH	Box dunk	No ingress	FM-24
B-04	BENCH	Harness pull and connector	No disconnect	FM-25
B-05	BENCH	Camera failure	Health reports fault; mission unaffected	FM-29
B-07	BENCH	Salvaged cell acceptance	Only compliant cells accepted and recorded	FM-23
B-06	BENCH	Clock without GNSS	Timestamps flagged until GNSS time	FM-30
R-01	REHEARSAL	Operator contingency rehearsal	All section 3.5 scenarios rehearsed	FM-42, FM-43, FM-44, FM-12
P-01	POOL	Internet absent	Templates offered	FM-38

6.7 What the rigs verify earlier

Item	First verified (Issue A)	Now also by	Still needs
FS-009 STOP ≤ 1 s	BENCH / POOL	L1-03 (to helm), L2-11 (to motors stopped)	On-water confirmation (POOL)
COM-003 telemetry and latency	POOL	L1-03, L1-05	Range effects (LAKE)
CAM-007 / MCP-D16 photo sync	POOL	L1-04	-
MCN-D21 link quality display	POOL	L1-07	-
MCN-D22 manual drive	POOL	L2 (motors follow the joystick)	Handling (POOL)
NAV-001 bidirectional thrust	POOL	L2-01, L3-01	-
PRP-D02 thrust ≥ 4 N	BENCH (estimate)	L3-01 (measured)	-
PRP-D03/D04 cruise throttle and power	POOL	L3-03 (estimate from data)	Confirm (POOL)
PRP-D11 / ENV-005 weed	LAKE	L3-02 (real weed)	Confirm (LAKE)
FS-005/006 stuck and shedding	POOL	SC-05/06 + L3-02	Confirm (POOL)
FS-001 / FS-004 on real hardware	SIM only	L2-09, L2-03/04	-
REC-D03 beacon patterns	LAKE	L2-12 (patterns)	Visibility (LAKE)
PWR-010 thermal	BENCH	L1-09 under a real workload	-

The FMEA check (SIM-D24) confirms that every failure mode with severity ≥ 8 has at least one test here.

7. Design constraints and budgets

- Same parameter file as the real helm, plus sitl.parm
- All timings in simulated time; runs faster than real time (5x used)
- Runs on a laptop or the Pi 5
- Rig L2 is run with props off, or in water with the tank covered. Removing the magnetic key is the emergency stop.

Budget	Allocation
Cost	£0 software (open source). Rig equipment, outside the Mk1 BOM: USB-serial adapter £3, load cell + HX711 £6, tub; bench power supply assumed owned.
LLM evaluation	API credit, capped per run
Mass / power	n/a

8. Verification and open items

Each requirement is verified at its stage using the method shown. Test procedures are written per stage and reference these IDs.

Stage	Test	Demo	Inspect	Analysis	Sim	Reqs
SIM	6	5	6	2	-	19
BENCH	7	2	2	-	-	11
POOL	-	-	-	-	-	0
LAKE	-	-	-	-	-	0

Ref	Open item
Model	Boat-model thrust, drag and inertia are estimates until the thrust stand and pool replace them (KCL section 8)
Slice 3	Mission Control against the simulator; C7 impostor-GCS detection (SC-37)
SIM-OI-1	Rig L2 needs the first parts order (FC, ESCs, motors, power parts). Rig L1 needs only a Pi Zero 2W and camera.